

Robust Hybrid Optimal Control of a Pressure-Fed Rocket Engine with Blowdown Dynamics, Gas Budget Constraints, and Binary Upstream Actuation

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1 Overview and Control Objective

We consider a pressure-fed rocket propulsion system with:

- One COPV pressurant source,
- Two propellant tanks (oxidizer and fuel),
- One upstream binary solenoid valve per tank,
- No downstream throttling authority,
- Continuous propellant outflow during engine operation.

The system exhibits *blowdown behavior*: when an upstream solenoid is closed, tank pressure decreases due to propellant depletion and ullage expansion.

1.1 Primary Control Objective

Given a desired thrust or altitude trajectory, determine the binary upstream valve commands

$$u_F(t), u_O(t) \in \{0, 1\}$$

that:

1. Track the desired thrust profile as closely as physically feasible,
2. Respect mixture ratio (O/F) and combustion stability constraints,
3. Respect COPV gas budget constraints,
4. Remain robust to modeling error and unmodeled losses,
5. Avoid instability due to blowdown dynamics.

2 System State Definition

2.1 Navigation State

$$x_v(t) = \begin{bmatrix} h(t) \\ v_z(t) \\ \theta(t) \end{bmatrix}$$

2.2 Pressurant and Tank State

Let:

$$x_p(t) = \begin{bmatrix} P_c(t) \\ P_r(t) \end{bmatrix}$$

where P_c is COPV pressure and P_r is regulator outlet pressure.

2.3 Tank Ullage and Feed Pressures

$$x_t(t) = \begin{bmatrix} P_{u,F}(t) \\ P_{u,O}(t) \\ P_{d,F}(t) \\ P_{d,O}(t) \end{bmatrix}$$

2.4 Complete State Vector

$$x(t) = \begin{bmatrix} x_v(t) \\ x_p(t) \\ x_t(t) \end{bmatrix} \in \mathbb{R}^9$$

3 Control Inputs

Binary upstream solenoids:

$$u(t) = \begin{bmatrix} u_F(t) \\ u_O(t) \end{bmatrix} \in \{0, 1\}^2$$

Implemented via PWM over short interval Δt .

4 Blowdown Ullage Dynamics

This section explicitly models the **pressure decay when valves are closed**.

4.1 Ullage Volume Evolution

Let tank $i \in \{F, O\}$ have:

$$V_{u,i}(t) = V_{u,i,0} + \int_0^t \frac{\dot{m}_i(\tau)}{\rho_i} d\tau$$

4.2 Ideal Gas Ullage Law

Assuming isothermal ullage:

$$P_{u,i}(t)V_{u,i}(t) = m_{u,i}(t)RT$$

4.3 Blowdown ODE (Valve Closed)

When $u_i = 0$:

$$\dot{P}_{u,i} = -\frac{P_{u,i}}{V_{u,i}} \dot{V}_{u,i} = -\frac{P_{u,i}}{V_{u,i}} \frac{\dot{m}_i}{\rho_i}$$

This term is *always negative* during firing and represents irreversible blowdown.

4.4 Pressurization ODE (Valve Open)

When $u_i = 1$:

$$\dot{P}_{u,i} = \alpha_i(P_r - P_{u,i}) - \frac{P_{u,i}}{V_{u,i}} \frac{\dot{m}_i}{\rho_i}$$

where α_i captures upstream valve + line conductance.

5 COPV Gas Dynamics and Budget Constraint

Assuming near-isothermal COPV:

$$m_c(t) = \frac{P_c(t)V_c}{RT_c}$$

5.1 Gas Consumption

$$\dot{m}_c = - \sum_{i \in \{F, O\}} u_i \dot{m}_{c \rightarrow i} - \dot{m}_{c, \text{loss}}$$

5.2 Hard Gas Constraint

$$m_c(t) \geq m_{c, \text{min}} \iff P_c(t) \geq P_{c, \text{min}}$$

This constraint is **inviolable**.

6 Injector, Chamber, and Thrust Model

6.1 Injector Mass Flow

$$\dot{m}_i = C_{d,i} A_i \sqrt{2\rho_i(P_{d,i} - P_{ch})}$$

6.2 Mixture Ratio

$$\text{MR} = \frac{\dot{m}_O}{\dot{m}_F}$$

6.3 Chamber Closure Equation

$$\dot{m}_O(P_{ch}) + \dot{m}_F(P_{ch}) = \frac{P_{ch} A_t}{c(P_{ch}, \text{MR})}$$

Solved each timestep by bracketed root finding.

6.4 Thrust

$$F = C_f(P_{ch}, \text{MR}) P_{ch} A_t$$

7 Hard Combustion Constraints

7.1 Mixture Ratio Bounds

$$\text{MR}_{\text{min}} \leq \text{MR}(t) \leq \text{MR}_{\text{max}}$$

7.2 Injector Stiffness

$$P_{d,i} - P_{ch} \geq \epsilon_i P_{ch}$$

7.3 Chugging Stability

$$\tau_{ch} = \frac{V_{ch}P_{ch}}{A_t c}, \quad f_{ch} = \frac{1}{2\pi\tau_{ch}}$$

Enforce:

$$\zeta(P_{ch}, L) \geq \zeta_{\min}$$

8 Outer Optimal Control (Thrust Reference)

8.1 Altitude-Based Thrust Command

$$a_z = k_p(h^{-h}) + k_d(v^{-v_z})$$

$$F_{\text{des}} = m \frac{a_z^{+g}}{\cos \theta}$$

8.2 Feasible Reference Projection

Solve:

$$\min_{F_{\text{ref}}} \int_0^T (F_{\text{ref}} - F_{\text{des}})^2 dt$$

subject to:

$$F_{\min}(t) \leq F_{\text{ref}} \leq F_{\max}(t)$$

$$\left| \dot{F}_{\text{ref}} \right| \leq \dot{F}_{\max}(t)$$

9 Robust Hybrid Optimal Control Problem

9.1 Discrete-Time Formulation

Let timestep k :

$$x_{k+1} = f_k(x_k, u_k) + Gw_k$$

where w_k represents bounded uncertainty.

9.2 Robust Min–Max MPC

$$\min_{u_{k:k+N-1}} \max_{w_{k:k+N-1} \in \mathcal{W}} \sum_{j=0}^{N-1} (\|F_j - F_{\text{ref},j}\|^2 + \lambda_g \Delta m_{c,j} + \lambda_s \|u_j - u_{j-1}\|_1)$$

9.3 Constraints (For All $w \in \mathcal{W}$)

$$m_c \geq m_{c,\min} \tag{1}$$

$$P_{u,i} \leq P_{u,\max} \tag{2}$$

$$\text{MR}_{\min} \leq \text{MR} \leq \text{MR}_{\max} \tag{3}$$

$$P_{d,i} - P_{ch} \geq \epsilon_i P_{ch} \tag{4}$$

9.4 Binary Nature

$$u_k \in \{0, 1\}^2$$

10 Enumerative Robust MPC Solution

At each timestep:

1. Enumerate admissible valve sequences (respecting min dwell time),
2. Propagate worst-case blowdown and gas loss,
3. Reject infeasible sequences,
4. Choose minimal worst-case cost.

This is tractable due to only two binary inputs.

11 Open/Close Decision Logic (Economic Interpretation)

Define marginal thrust benefit:

$$\Delta F_i = F(u_i = 1) - F(u_i = 0)$$

Define marginal gas cost:

$$\Delta m_{c,i} = m_c(u_i = 0) - m_c(u_i = 1)$$

Valve i opens if:

$$q_F(F_{\text{ref}} - F)\Delta F_i \geq \lambda_g \Delta m_{c,i} + \lambda_s$$

and constraints remain feasible under worst-case blowdown.

12 Supervisory Cutoff

Predict apogee:

$$h_{\text{ap}} = h + \frac{v_z^2}{2g}$$

If:

$$h_{\text{ap}} \geq h^* - \delta \Rightarrow u_F = u_O = 0$$

13 Conclusion

This document presents a complete, mathematically closed, robust optimal control framework for thrust shaping in a pressure-fed rocket engine with:

- explicit blowdown modeling,
- binary upstream actuation,
- hard gas constraints,

- robust min–max MPC,
- combustion stability enforcement.

No assumptions are implicit. All dynamics, constraints, and decision rules are explicitly stated.